



AISSMS

COLLEGE OF ENGINEERING

ज्ञानम् सकलजनहिताय

An Autonomous Institute Affiliated with
Savitribai Phule Pune University

Accredited by NAAC with "A+" Grade | NBA - 7 UG Programmes



A.I.S.S.M.S. COLLEGE OF ENGINEERING PUNE 411001

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Expt. No	Title of the Experiment	Date on which		Page No.	Remarks and Signature
		Exp. Per	Exp. Sub.		
410250 : High Performance Computing					
Any 4 Assignments and 1 Mini Project are Mandatory					
Group 1					
1	Design and implement Parallel Breadth First Search and Depth First Search based on existing algorithms using OpenMP. Use a Tree or an undirected graph for BFS and DFS				
2	Write a program to implement Parallel Bubble Sort and Merge sort using OpenMP. Use existing algorithms and measure the performance of sequential and parallel algorithms				
3	Implement Min, Max, Sum and Average operations using Parallel Reduction				
4	Write a CUDA Program for: 1. Addition of two large vectors 2. Matrix Multiplication using CUDA C				
5	Implement HPC application for AI/ML domain.				
Group 2					
6	Mini Project: Evaluate performance enhancement of parallel Quicksort Algorithm using MPI				
7	Mini Project: Implement Huffman Encoding on GPU				
8	Mini Project: Implement Parallelization of Database Query optimization				
9	Mini Project: Implement Non-Serial Polyadic Dynamic Programming with GPU Parallelization				



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410251 : Deep Learning				
Any 3 Assignments and 1 Mini Project are Mandatory				
Group 1				
1	Linear regression by using Deep Neural network: Implement Boston housing price prediction problem by Linear regression using Deep Neural network. Use Boston House price prediction dataset.			
2	Classification using Deep neural network (Any One from the following) 1. Multiclass classification using Deep Neural Networks: Example: Use the OCR letter recognition dataset https://archive.ics.uci.edu/ml/datasets/letter+recognition 2. Binary classification using Deep Neural Networks Example: Classify movie reviews into positive" reviews and "negative" reviews, just based on the text content of the reviews. Use IMDB dataset			
3	Convolutional neural network (CNN) (Any One from the following) • Use any dataset of plant disease and design a plant disease detection system using CNN. • Use MNIST Fashion Dataset and create a classifier to classify fashion clothing into categories			
4	Recurrent neural network (RNN) Use the Google stock prices dataset and design a time series analysis and prediction system using RNN			
Group 2				
5	Mini Project: Human Face Recognition			
6	Mini Project: Gender and Age Detection: predict if a person is a male or female and also their age			
7	Mini Project: Colorizing Old B&W Images: colour old black and white images to colourful images			

This is to certify that **Miss. Piyusha Rajendra Supe** of class **BE-B (Computer)** Roll No. **23CO315** has completed all the practical work as listed above, satisfactorily in the subject of **High Performance Computing** and **Deep Learning** under **Laboratory Practice V - 410254** in the Department of **Computer Engineering** as prescribed by the Savitribai Phule Pune University. During the academic year **2025-2026**

Date:

Prof. S. S. Jadhav
(Faculty In-charge)

Dr. S. V. Athawale
(Head of Department)

Piyusha Supe (23CO315)



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410252(A): Natural Language Processing					
Any 5 Assignments and 1 Mini Project are mandatory					
Group 1					
1	Perform tokenization (Whitespace, Punctuation-based, Treebank, Tweet, MWE) using NLTK library. Use porter stemmer and snowball stemmer for stemming. Use any technique for lemmatization. Input / Dataset –use any sample sentence				
2	Perform bag-of-words approach (count occurrence, normalized count occurrence), TF-IDF on data. Create embeddings using Word2Vec. Dataset to be used: https://www.kaggle.com/datasets/CooperUnion/cardataset				
3	Perform text cleaning, perform lemmatization (any method), remove stop words (any method), label encoding. Create representations using TF-IDF. Save outputs. Dataset: https://github.com/PICT-NLP/BE-NLP-Elective/blob/main/3-Preprocessing/News_dataset.pickle				
4	Create a transformer from scratch using the Pytorch library				
5	Morphology is the study of the way words are built up from smaller meaning bearing units. Study and understand the concepts of morphology by the use of add delete table				
Group 2					
6	Mini Project (Fine tune transformers on your preferred task) Finetune a pretrained transformer for any of the following tasks on any relevant dataset of your choice: • Neural Machine Translation • Classification • Summarization				
7	Mini Project - POS Taggers For Indian Languages				
8	Mini Project -Feature Extraction using seven moment variants				
9	Mini Project -Feature Extraction using Zernike Moments				



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410253(C) : Business Intelligence

Any 5 Assignments and 1 Mini Project are mandatory

Group 1

1	Import the legacy data from different sources such as (Excel, Sql Server, Oracle etc.) and load in the target system. (You can download sample database such as Adventure works, Northwind, foodmart etc.)				
2	Perform the Extraction Transformation and Loading (ETL) process to construct the database in the Sql server.				
3	Create the cube with suitable dimension and fact tables based on ROLAP, MOLAP and HOLAP model.				
4	Import the data warehouse data in Microsoft Excel and create the Pivot table and Pivot Chart				
5	Perform the data classification using classification algorithm. Or Perform the data clustering using clustering algorithm.				

Group 2

6	Mini Project: Each group of 4 Students (max) assigned one case study for this; A BI report must be prepared outlining the following steps: a) Problem definition, identifying which data mining task is needed. b) Identify and use a standard data mining dataset available for the problem				
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This is to certify that **Miss. Piyusha Rajendra Supe** of class **BE-B (Computer)** Roll No. **23CO315** has completed all the practical work as listed above, satisfactorily in the subject of **Natural Language Processing** and **Business Intelligence** under **Laboratory Practice VI - 410255** in the Department of **Computer Engineering** as prescribed by the Savitribai Phule Pune University. During the academic year **2025-2026**

Date:

Prof. S. J. Pachouly
(Faculty In-charge)

Dr. S. V. Athawale
(Head of Department)

Piyusha Supe (23CO315)